Tutorial Questions

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Question 1

A vacuum gage conncected to a tank reads 30 kPa at a location where the barometric reading is 755 mmHg. Determine the absolute pressure in the tank. Take $\rho_{Hg} = 13{,}590 \ \ \text{kg/m}^3$.

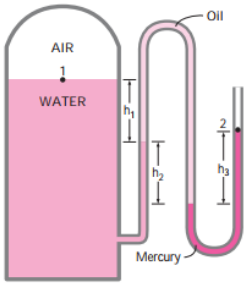
Solution

The vacuum pressure is recorded at 30 kPa, and the atmospheric pressure reading is 755 mmHg. The absolute pressure is then found by subtracting the vacuum pressure from the atmospheric pressure.

$$P_{abs} = P_{atm} - P_{vac}$$
$$P_{abs} = (13590 \times 9.81 \times 0.755) - (30 \times 10^3)$$
$$\implies 70.66 \text{ kPa}$$

Question 2

The water in a tank is pressurized by air, and the pressure is measured by a multifluid manometer as shown in the figure below. Determine the gage pressure of air in the tank if $h_1 = 0.2$ m, $h_2 = 0.3$ m, and $h_3 = 0.46$ m. Take the densities of water, oil, and mercury to be 1000 kg/m^3, 850 kg/m^3, and 13,600 kg/m^3, respectively.



Solution

Since we are told to find the gage pressure of the air in the tank, the atmospheric pressure is neglected.

$$\begin{gathered} P_1 + [\rho g h_1]_w + [\rho g h_2]_{oil} - [\rho g h_3]_{Hg} = 0 \\ P_1 = [\rho g h_3]_{Hg} - [\rho g h_2]_{oil} - [\rho g h_1]_w \\ \implies 9.81 \times [(13600 \times 0.46) - (850 \times 0.3) - (1000 \times 0.2)] \\ \implies 56.91 \text{ kPa} \end{gathered}$$

Question 3

The gage pressure in a liquid at a depth of 3 m is read to be 28 kPa. Determine the gage pressure in the same liquid at a depth of 12 m.

Solution

$$\begin{gathered} P_1 = \rho g h_1 \\ P_2 = \rho g h_2 \\ \frac{P_1}{P_2} = \frac{h_1}{h_2} \\ P_2 = \frac{h_2}{h_1} \times P_1 = \frac{12}{3} \times 28 = 112 \text{ kPa} \end{gathered}$$

Question 4

The barometer of a mountain hiker reads 930 mbars at the beginning of a hiking trip and 780 mbars at the end. Neglecting the effect of altitude on local gravitational acceleration, determine the vertical distance climbed. Assume an average air density of 1.20 kg/m^3.

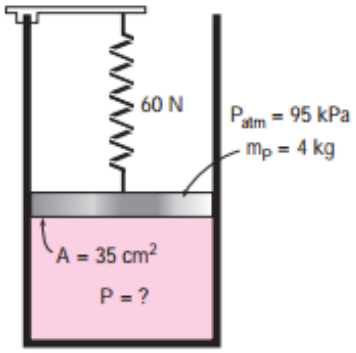
Solution

$$\begin{gathered} \Delta P = \rho g h \\ (930 - 780) \times 10^2 = 1.20 \times 9.81 \times h \\ h = \frac{15000}{1.20 \times 9.81} = 1274.2 \text{ m} \end{gathered}$$

Question 5

A gas is contained in a vertical, frictionless piston–cylinder device. The piston has a mass of 4 kg and a crosssectional area of 35 cm^2. A compressed spring

above the piston exerts a force of 60 N on the piston. If the atmospheric pressure is 95 kPa, determine the pressure inside the cylinder.



Solution

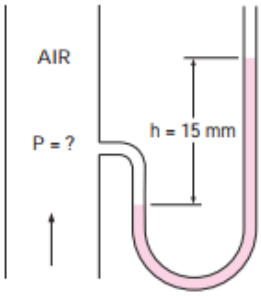
The absolute pressure inside the cylinder will be the sum of three pressures; from the spring, the weight of the piston and the atmospheric pressure
Total pressure exerted = [atmospheric pressure] + [pressure from spring] + [pressure from weight]

$$P = 95000 + \left[\frac{60}{35 \times 10^{-4}}\right] + \left[\frac{4 \times 9.81}{35 \times 10^{-4}}\right] = 123.354 \text{ kPa}$$

Question 6

A mercury manometer ($\rho = 13{,}600$ kg/m^3) is connected to an air duct to measure the pressure inside. The difference in the manometer levels is 15 mm, and the atmospheric pressure is 100 kPa.

1. Judging from the figure, determine if the pressure in the duct is above or below the atmospheric pressure.
2. Determine the absolute pressure in the duct.



Solution

1. The pressure in the duct is above the atmospheric pressure.
2. Using the equations to find pressure in a manometer, we get;

$$P - [\rho g h]_{Hg} = P_{atm}$$
$$P = P_{atm} + [\rho g h]_{Hg} = 100000 + (13600 \times 9.81 \times 0.015)$$
$$P = 102 \text{ kPa}$$

Question 7

The maximum blood pressure in the upper arm of a healthy person is about 120 mmHg. If a vertical tube open to the atmosphere is connected to the vein in the arm of the person, determine how high the blood will rise in the tube. Take the density of the blood to be 1050 kg/m^3.

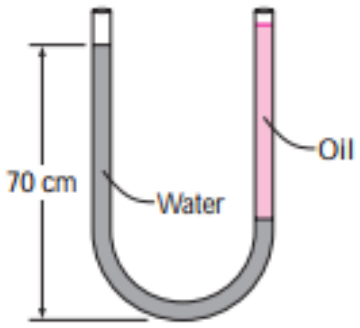
Solution

The blood pressure in the upper arm of a healthy person is recorded at 120 mmHg. When connected to a vertical tube, it will exert that same pressure, leading to a corresponding rise in the blood in the tube.

$$[\rho g h]_{Hg} = [\rho g h]_{blood}$$
$$h_{blood} = \frac{[\rho h]_{Hg}}{\rho_{blood}} = \frac{13600 \times 0.12}{1050} = 1.55 \text{ m}$$

Question 8

Consider a U-tube whose arms are open to the atmosphere. Now water is poured into the U-tube from one arm, and light oil ($\rho = 790$ kg/m^3) from the other. One arm contains 70-cm-high water, while the other arm contains both fluids with an oil-to-water height ratio of 6. Determine the height of each fluid in that arm.



Solution

The pressure exerted on the bottom of the tube by both sides are the same, since they are exposed to the same atmosphere. Hence, $[\rho gh]_w = [\rho g(6x)]_{oil} + [\rho gx]_w$
Whereby;
h is the height of the water in the first column, 70 cm
x is the height of the water in the second column.

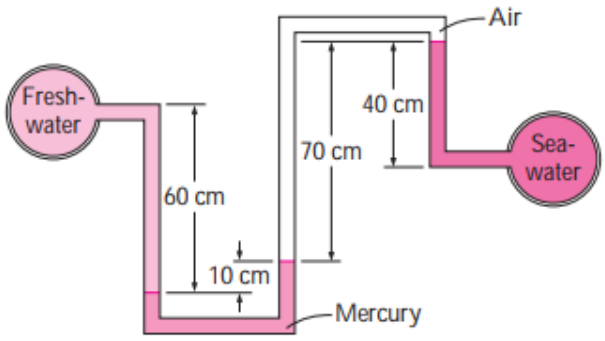
Making x the subject in this case, we get;

$$x = \frac{[\rho h]_w}{(6\rho_{oil}) + \rho_w}$$
$$x = \frac{1000 \times 70}{(6 \times 790) + 1000}$$
$$\implies 12.2 \text{ cm}$$

The height of the oil is six times the height of water, making that 73.2 cm.

Question 9

Freshwater and seawater flowing in parallel horizontal pipelines are connected to each other by a double U-tube manometer, as shown in the figure below. Determine the pressure difference between the two pipelines. Take the density of seawater at that location to be $\rho = 1035$ kg/m^3. Can the air column be ignored in the analysis?



Solution

Since the air column has a negligible density ($\rho_{air} = 1.20$ kg/m^3), the pressure exerted by the air can be neglected.
Using this knowledge we can find our pressure difference with this equation;

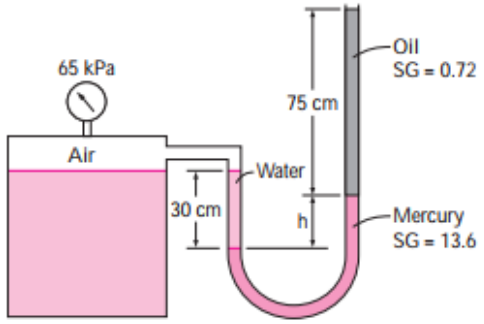
$$P_1 + [\rho gh]_{fresh} - [\rho gh]_{Hg} + [\rho gh]_{sea} = P_2$$

The pressure difference between the two pipelines is then;

$$\begin{aligned} P_1 - P_2 &= [\rho gh]_{Hg} - [\rho gh]_{fresh} - [\rho gh]_{sea} \\ \implies & 9.81[(13600 \times 0.1) - (1000 \times 0.6) - (1035 \times 0.4)] \\ \implies & 3.39 \text{ kPa} \end{aligned}$$

Question 10

The gage pressure of the air in the tank shown in the figure below is measured to be 65 kPa. Determine the differential height h of the mercury column.



Solution

$$\begin{aligned} P_{air} + [\rho gh]_w - [\rho gh]_{Hg} - [\rho gh]_{oil} &= P_{atm} \\ h_{Hg} &= \frac{[P_{air} - P_{atm}] + [\rho gh]_w - [\rho gh]_{oil}}{\rho_{Hg} \times g} \\ \implies & \frac{(65 \times 10^3) + (1000 \times 9.81 \times 0.3) - (720 \times 9.81 \times 0.75)}{13600 \times 9.81} \\ \implies & 0.4696 \text{ m} = 46.96 \text{ cm} \end{aligned}$$